Figure 1.6 shows a network of three nodes connected by a network. The network facilitates broadcast of the

- transactions initiated by users
- blocks formed out of the transactions

These transactions and blocks constitute the payload of the network and, eventually, after verification and validation, are recorded on the distributed ledger.

A network identifier identifies a blockchain network of nodes. Network ID #1, for example, is the main Ethereum public network; network ID #4 is a public network called Rinkeby (https://www.rinkeby.io), and so on. You'll have to indicate the network by using its identifier while deploying your smart contract on the network. The participants on a given network will share a unified distributed ledger for recording their transaction details.

The smart contracts are deployed in a sandbox environment such as a virtual machine (VM) hosted by a blockchain node. The syntax of a smart contract is similar to a class in an OO (object-oriented) language. It contains data, functions, and rules for the execution of functions. Calling or invoking a smart contract function generates the transactions that are recorded on the blockchain, as shown in figure 1.7. If any of the verification and validation rules fails, the function invocation is reverted. But if the execution is successful, the generated transactions (Txs) are broadcast to the network for recording, as shown in figure 1.7. The figure illustrates how a function call is transformed into actions that are recorded on the blockchain.